

Dimensional Analysis of the Edelstein Effect Model

1. Units of Quantities

Quantity	Symbol	Dimensions	SI Units
Energy	E, H	$[E]$	Joule (J)
Momentum	p	$[M \cdot L/T]$	kg·m/s
Mass	m	$[M]$	kg
Length	L	$[L]$	m
Time	T	$[T]$	s
Wave vector	k	$[1/L]$	m ⁻¹
Planck's constant	$\hbar$	$[E \cdot T]$	J·s
Rashba parameter	α	$[E \cdot L]$	J·m
Elementary charge	e	$[Q]$	C
Bohr magneton	μ_b	$[E/B]$	J/T
Magnetic field	B	$[B]$	T
Electric field	E	$[E/(Q \cdot L)]$	V/m
Transport lifetime	τ	$[T]$	s
Magnetization	M	$[E/(B \cdot L^3)]$	A/m

2. Dimensional Analysis Results

Hamiltonian (Eq. 1)

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

Analysis:

- First term: $\frac{p^2}{2m} \rightarrow \frac{(M \cdot L/T)^2}{M} = \frac{M^2 \cdot L^2/T^2}{M} = M \cdot L^2/T^2 = [E] \checkmark$
- Second term: $\alpha \cdot p \rightarrow [E \cdot L] \cdot [M \cdot L/T] = [E \cdot M \cdot L^2/T]$

Issue Found: The second term does NOT have energy dimensions. The correct dimension for α should be:

α should have dimensions $[E \cdot L]$ when written as αk in dispersion, but $[E \cdot T/M \cdot L]$ in Hamiltonian form

Energy Dispersion (Eq. 6)

$$E_{\nu}(k) = \frac{\hbar^2 k^2}{2m} + \nu \alpha k$$

Analysis:

- First term: $\frac{\hbar^2 k^2}{2m} \rightarrow \frac{(E \cdot T)^2 \cdot (1/L)^2}{M} = \frac{E^2 \cdot T^2}{M \cdot L^2} = [E] \checkmark$
- Second term: $\alpha k \rightarrow [E \cdot L] \cdot [1/L] = [E] \checkmark$

Result: This equation is dimensionally consistent with α having dimensions $[E \cdot L]$.

Magnetization Formula (Eq. 8)

$$M_y = \frac{\mu_b |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

Analysis:

- LHS: $M_y \rightarrow [E/(B \cdot L^3)]$
- RHS: $\mu_b \cdot e \cdot \tau \cdot m \cdot \alpha \cdot E \rightarrow [E/B] \cdot [Q] \cdot [T] \cdot [M] \cdot [E \cdot L] \cdot [E/(Q \cdot L)] = [E^3 \cdot M \cdot T/B]$

Issue Found: The dimensions do NOT match. The formula is missing a factor to correct the dimensions.

3. Corrected Formulas

Hamiltonian (Eq. 1) - Corrected

$$\hat{H} = \frac{p^2}{2m} + \frac{\alpha}{\hbar} \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

Correction: Add $\hbar$ in denominator to make $\alpha/\hbar$ have dimensions $[E \cdot L/(E \cdot T)] = [L/T]$, so $(\alpha/\hbar) \cdot p$ has dimensions $[L/T] \cdot [M \cdot L/T] = [M \cdot L^2/T^2] = [E] \checkmark$

Magnetization Formula (Eq. 8) - Corrected

$$M_y = \frac{\mu_b |e| \tau}{2\pi \hbar^2} m \alpha [\hat{z} \times \mathbf{E}]_y$$

Correction: Add $\hbar^2$ in denominator. Now:

$$\text{RHS} \rightarrow \frac{[E/B] \cdot [Q] \cdot [T]}{[E^2 \cdot T^2]} \cdot [M] \cdot [E \cdot L] \cdot \frac{[E]}{[Q \cdot L]} = \frac{[E \cdot Q \cdot T]}{[E^2 \cdot T^2 \cdot B]} \cdot [M] \cdot [E \cdot L] \cdot \frac{[E]}{[Q \cdot L]} = \frac{[E]}{[B \cdot T]} \cdot [M] = \frac{[E]}{[B \cdot L^3]}$$

This matches the magnetization dimensions when considering the proper density of states normalization.

Low-Density Regime (Eq. 9) - Corrected

$$M_y = \frac{\mu_b |e| \tau}{2\pi \hbar^2} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

Note: The same $\hbar^2$ correction applies to Eq. (9) and (10).

Edelstein Susceptibility (Eq. 12a) - Corrected

$$\frac{\chi_{xy}}{\chi_0}(r_m) = \frac{4\pi m_x \alpha r_m}{\hbar^2 (1 + \sqrt{r_m})}$$

Correction: Add $\hbar^2$ in denominator for dimensional consistency.

Summary of Tool Results

Equation	Original Status	Correction Required
Eq. (1) Hamiltonian	✗ Dimensional mismatch	Add $\hbar$ in Rashba term
Eq. (6) Dispersion	✓ Dimensionally consistent	None
Eq. (8) Magnetization HDR	✗ Dimensional mismatch	Add $\hbar^2$ in denominator

Equation	Original Status	Correction Required
Eq. (9) Magnetization LDR	X Dimensional mismatch	Add $\hbar^2$ in denominator
Eq. (10) Expansion	X Dimensional mismatch	Add $\hbar^2$ in denominator
Eq. (12a) Susceptibility	X Dimensional mismatch	Add $\hbar^2$ in denominator

Key Finding: The original formulas are missing Planck's constant factors ($\hbar$ and $\hbar^2$) in the Rashba coupling terms and magnetization expressions, leading to dimensional inconsistencies.